

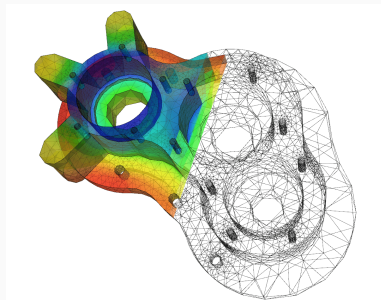
Finite element method

Linear elastodynamics

ME473 Dynamic finite element analysis of structures

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2025



Where do we stand?

Week	Module	Lecture topic	Mini-projects
1	Linear elastodynamics	Strong and weak forms	
2		Galerkin and FE methods	
3		Systematisation of the procedure	Groups formation

Summary

- Recap week 2
- FEM in global coordinate system
 - 1D and 2D elements and shape functions
 - Example: longitudinal vibration of a bar
- FEM in local coordinate system
 - Localization and elementary quantities
 - Automating integration and archetypal shape functions
 - Example: dynamic analysis of a clamped beam

Recommended readings

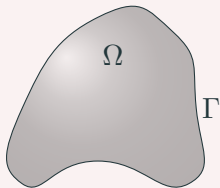
- ① Gmür, Dynamique des structures (§3.1, §3.2 and §3.3)
- ② Neto et al., Engineering Computation of Structures (§2.3 and Chapter 7)

Recap week 2

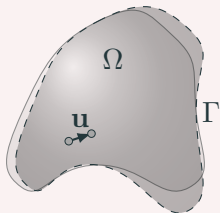
Discretisation with Galerkin method

Formulations of elastodynamics

Undeformed solid



Deformed solid



Strong form

$$\nabla^T \mathbf{C} \nabla \mathbf{u} + \mathbf{f} = \rho \ddot{\mathbf{u}}$$



Weak form

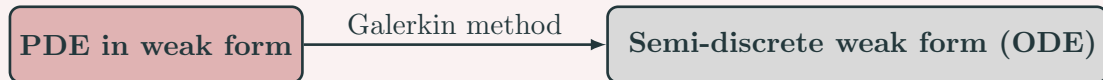
$$\begin{aligned} \int_{\Omega} (\nabla \delta \mathbf{u})^T \mathbf{C} \nabla \mathbf{u} d\Omega + \int_{\Omega} \rho \delta \mathbf{u}^T \ddot{\mathbf{u}} d\Omega \\ = \int_{\Gamma_{\sigma}} \delta \mathbf{u}^T \hat{\mathbf{f}} d\Gamma + \int_{\Omega} \delta \mathbf{u}^T \mathbf{f} d\Omega \end{aligned}$$



Semi-discrete weak form

$$\mathbf{M} \ddot{\mathbf{q}}(t) + \mathbf{K} \mathbf{q}(t) = \mathbf{r}(t)$$

Galerkin method



Shape functions:

$$\mathbf{u}^h(\mathbf{x}, t) = \mathbf{H}(\mathbf{x})\mathbf{q}(t) \quad \text{and} \quad \delta \mathbf{u}^h(\mathbf{x}) = \mathbf{H}(\mathbf{x})\delta \mathbf{q}.$$

$$\begin{pmatrix} u_1^h(\mathbf{x}, t) \\ u_2^h(\mathbf{x}, t) \\ u_3^h(\mathbf{x}, t) \end{pmatrix} = \begin{bmatrix} h_{11}(\mathbf{x}) & h_{12}(\mathbf{x}) & \dots & h_{1n}(\mathbf{x}) \\ h_{21}(\mathbf{x}) & h_{22}(\mathbf{x}) & \dots & h_{2n}(\mathbf{x}) \\ h_{31}(\mathbf{x}) & h_{32}(\mathbf{x}) & \dots & h_{3n}(\mathbf{x}) \end{bmatrix} \begin{pmatrix} q_1(t) \\ q_2(t) \\ \vdots \\ q_n(t) \end{pmatrix}$$

$$\begin{pmatrix} \delta u_1^h(\mathbf{x}) \\ \delta u_2^h(\mathbf{x}) \\ \delta u_3^h(\mathbf{x}) \end{pmatrix} = \begin{bmatrix} h_{11}(\mathbf{x}) & h_{12}(\mathbf{x}) & \dots & h_{1n}(\mathbf{x}) \\ h_{21}(\mathbf{x}) & h_{22}(\mathbf{x}) & \dots & h_{2n}(\mathbf{x}) \\ h_{31}(\mathbf{x}) & h_{32}(\mathbf{x}) & \dots & h_{3n}(\mathbf{x}) \end{bmatrix} \begin{pmatrix} \delta q_1 \\ \delta q_2 \\ \vdots \\ \delta q_n \end{pmatrix}$$

Definitions

- **Stiffness matrix** ($n \times n$):

$$\mathbf{K} = \int_{\Omega} \mathbf{B}^T \mathbf{C} \mathbf{B} d\Omega$$

$\mathbf{B}$ is the ($6 \times n$) deformation matrix : $\mathbf{B} = \nabla \mathbf{H}$.

- **Mass matrix** ($n \times n$):

$$\mathbf{M} = \int_{\Omega} \rho \mathbf{H}^T \mathbf{H} d\Omega.$$

- **Applied forces vector** ($n \times 1$):

$$\mathbf{r}(t) = \int_{\Gamma_{\sigma}} \mathbf{H}^T \hat{\mathbf{f}} d\Gamma + \int_{\Omega} \mathbf{H}^T \mathbf{f} d\Omega.$$

$$\nabla = \begin{bmatrix} \partial_x & 0 & 0 \\ 0 & \partial_y & 0 \\ 0 & 0 & \partial_z \\ 0 & \partial_z & \partial_y \\ \partial_z & 0 & \partial_x \\ \partial_y & \partial_x & 0 \end{bmatrix}$$

Advantages and drawbacks of Galerkin method

Advantages:

- ✓ Converges quickly with appropriate shape functions.
- ✓ Provides a systematic and structured approach for approximating solutions.
- ✓ The same set of functions is used to express real and virtual variables.

Drawbacks:

- ✗ Accuracy heavily dependent on choice of basis functions.
- ✗ No physical interpretation of the unknown variable $\mathbf{q}(t)$.
- ✗ The formulation of initial and boundary conditions in the discretized form is cumbersome.

Displacements approximation in finite element method

Let p the number of nodes of the mesh.

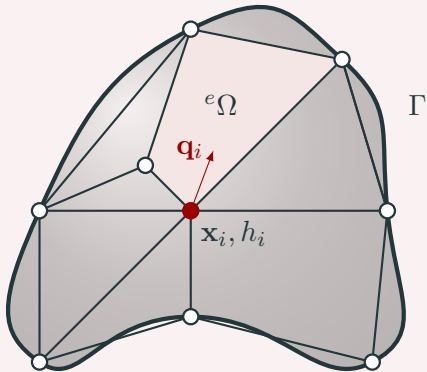
$$\mathbf{u}^h(\mathbf{x}, t) = \mathbf{H}(\mathbf{x})\mathbf{q}(t) = \sum_{i=1}^p h_i(\mathbf{x})\mathbf{q}_i(t)$$

■ $\mathbf{H}(\mathbf{x})$ is a $3 \times 3p$ matrix of **shape functions**:

$$\mathbf{H} = [h_1\mathbf{I} \mid h_2\mathbf{I} \mid \dots \mid h_i\mathbf{I} \mid \dots \mid h_p\mathbf{I}]$$

$\mathbf{I}$ is the 3×3 identity matrix.

■ $\mathbf{q}(t)$ is a $3p \times 1$ vector of (*unknown*) **nodal displacements**.



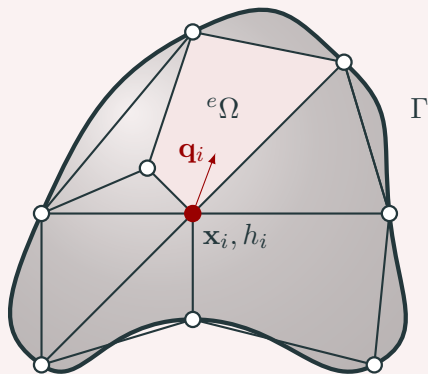
Global nodal shape functions requirements

Properties of h_i :

- Linearly independent polynomial basis.
- Satisfy Kronecker delta property:

$$h_i(\mathbf{x}_i) = 1 \quad \text{and} \quad h_i(\mathbf{x}_j) = 0.$$

- Vanish on non-adjacent elements.
- Continuous at interfaces.
- Differentiable inside elements.
- Ensure rigid body motion & constant deformations.



$$\sum_{i=1}^p h_i(\mathbf{x}) = 1 \quad \text{and} \quad \sum_{i=1}^p \nabla h_i(\mathbf{x}) = \mathbf{0}.$$

Types of analysis

Static analysis

Constant loads

Determines the structural deformation $\mathbf{q}$ due to constant loads.

$$\mathbf{K}\mathbf{q} = \mathbf{r}$$

- No inertia effects
- No time dependence
- Equilibrium between internal and external forces

Modal analysis

Natural frequencies

Studies the intrinsic dynamic properties of the structure.

$$\mathbf{M}\ddot{\mathbf{q}}(t) + \mathbf{K}\mathbf{q}(t) = \mathbf{0}$$

- No external excitation
- Resonance when excitation frequency $\approx$ natural frequency

Transient analysis

Time-dependent loads

Determines the structural response $\mathbf{q}(t)$ to time-dependent excitations

$$\mathbf{M}\ddot{\mathbf{q}}(t) + \mathbf{K}\mathbf{q}(t) = \mathbf{r}(t)$$

- Inertia effects included
- Response evolves in time
- Used for impacts, vibrations and dynamic loading

Modal analysis

- Free vibrations: $\mathbf{f} = 0$, thus $\mathbf{r}(t) = \mathbf{0}$.
- Assume harmonic solution:

$$\mathbf{q}(t) = \mathbf{p}\alpha \cos(\omega t - \varphi)$$

- $\mathbf{p}$ is the mode shape (eigenvector).
 - ω is the natural angular frequency (eigenvalue) [rad/s].
 - Regular frequency $f = \omega/2\pi$ [Hz].
 - α (amplitude) and φ (phase) depend on initial conditions.
- Leads to the eigenvalue problem:

$$(\mathbf{K} - \omega^2\mathbf{M})\mathbf{p} = \mathbf{0}.$$

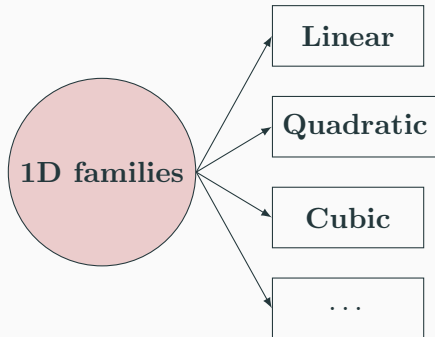
- Non-trivial solutions exist only if:

$$\det(\mathbf{K} - \omega^2\mathbf{M}) = 0.$$

The eigenvalues $\lambda_i = \omega_i^2$ are the squares of the natural angular frequencies.
The eigenvectors $\mathbf{p}_i$ are the corresponding mode shapes.

1D and 2D elements and global shape functions

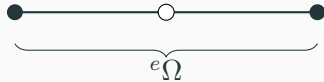
Shape functions for one dimensional structures



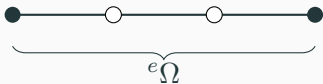
1D finite elements



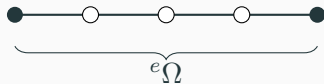
(a) Linear element



(b) Quadratic element

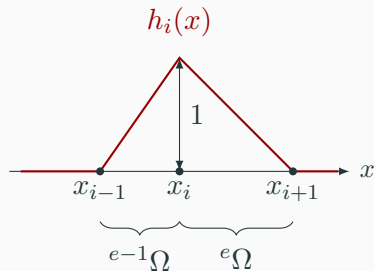


(c) Cubic element



(d) Quartic element

Linear shape functions



$$h_i(x) = \begin{cases} l_2[x_{i-1}, x_i](x) = \frac{x - x_{i-1}}{x_i - x_{i-1}} & x \in e^{-1}\Omega \\ l_1[x_i, x_{i+1}](x) = \frac{x - x_{i+1}}{x_i - x_{i+1}} & x \in e\Omega \\ 0 & \text{otherwise} \end{cases}$$

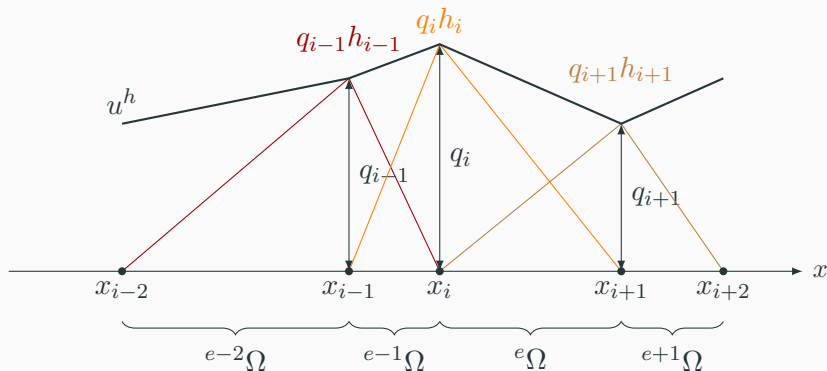
- h_i are piecewise linear functions that, within each finite element, corresponds to a first-degree Lagrange polynomial:

$$l_j[x_1, \dots, x_n](x) = \prod_{\substack{m=1 \\ m \neq j}}^n \frac{x - x_m}{x_j - x_m}$$

- Any linear piecewise function can be expressed as a linear combination of the h_i .

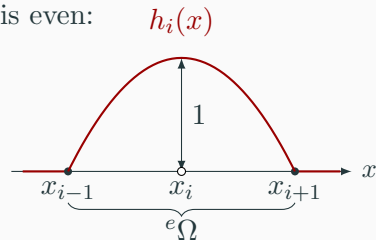
1D displacement approximation via linear shape functions

$$u^h(x, t) = \sum_{i=1}^p h_i(x) q_i(t)$$



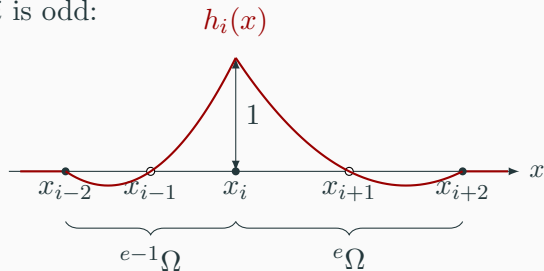
Quadratic shape functions

If i is even:



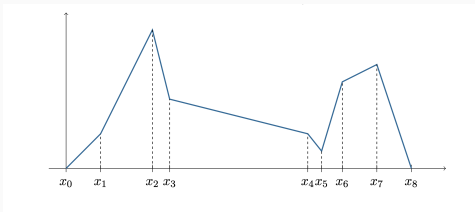
$$h_i(x) = \begin{cases} l_2[x_{i-1}, x_i, x_{i+1}](x) & x \in {}^e\Omega \\ 0 & \text{otherwise} \end{cases}$$

If i is odd:

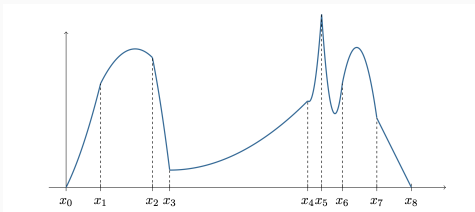


$$h_i(x) = \begin{cases} l_3[x_{i-2}, x_{i-1}, x_i](x) & x \in {}^{e-1}\Omega \\ l_1[x_i, x_{i+1}, x_{i+2}](x) & x \in {}^e\Omega \\ 0 & \text{otherwise} \end{cases}$$

Examples of 1D displacement approximation

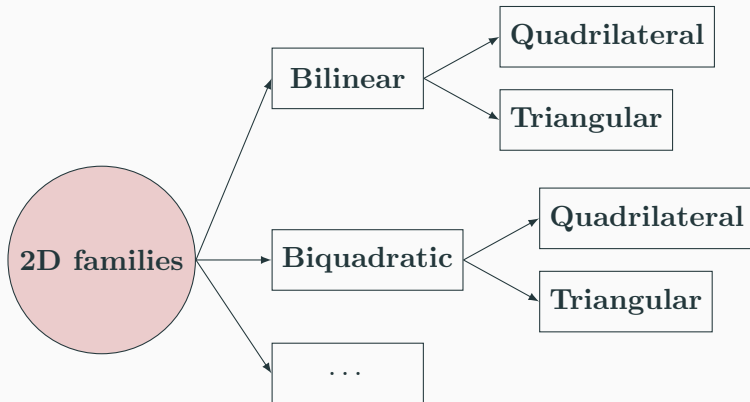


Displacement approximation via linear shape functions



Displacement approximation via quadratic shape functions

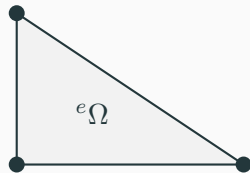
Shape functions for two-dimensional structures



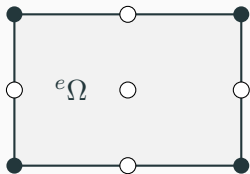
2D finite elements



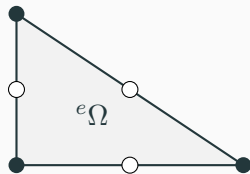
(a) Bilinear Quadrilateral



(b) Bilinear Triangular

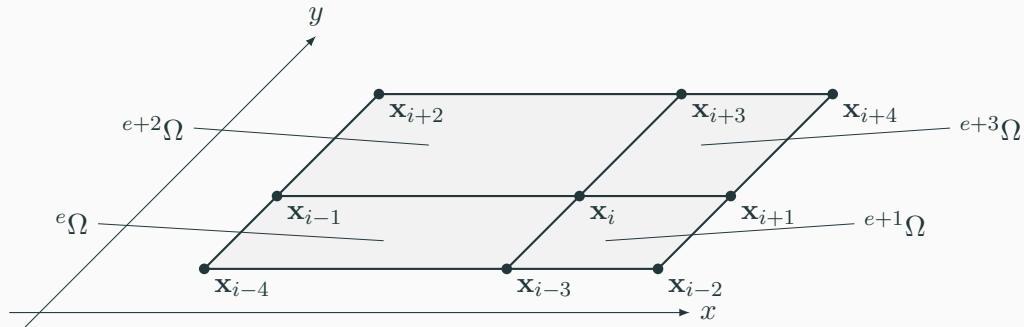


(c) Biquadratic Quadrilateral

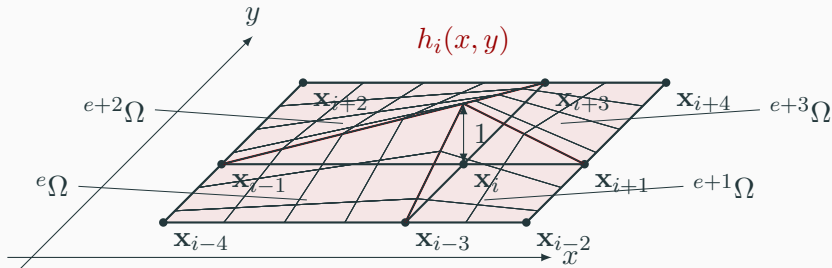


(d) Biquadratic Triangular

Bilinear quadrilateral finite elements mesh

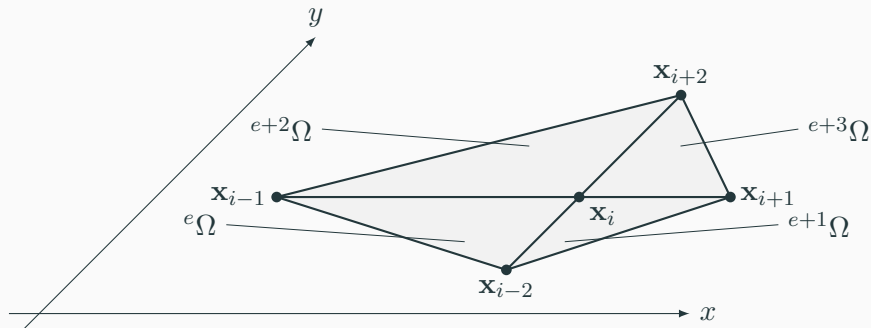


Bilinear quadrilateral shape functions

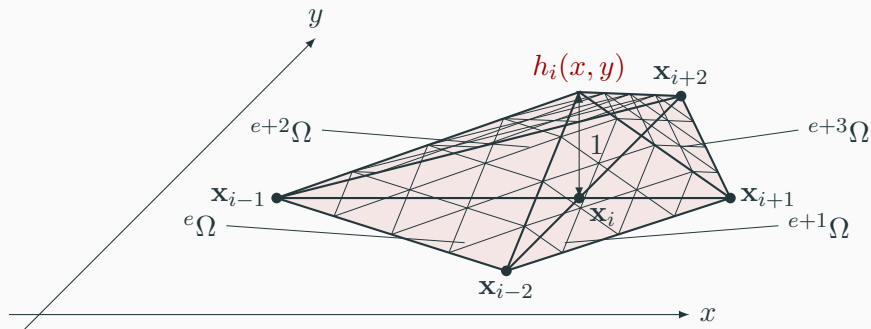


$$h_i(x, y) = \begin{cases} l_2[x_{i-1}, x_i](x)l_2[y_{i-3}, y_i](y) & x \in e\Omega \\ l_1[x_i, x_{i+1}](x)l_2[y_{i-3}, y_i](y) & x \in e+1\Omega \\ l_2[x_{i-1}, x_i](x)l_1[y_i, y_{i+3}](y) & x \in e+2\Omega \\ l_1[x_i, x_{i+1}](x)l_1[y_i, y_{i+3}](y) & x \in e+3\Omega \\ 0 & \text{otherwise} \end{cases}$$

Bilinear triangular finite elements mesh

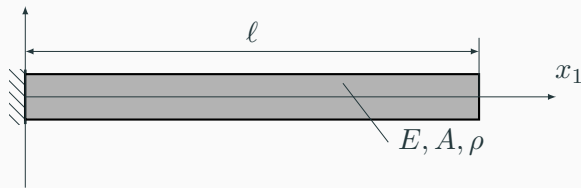


Bilinear triangular shape functions



Example: longitudinal vibration of a bar

Example - FE approximation of longitudinal vibrations of a bar



- A cross-sectional area
- ℓ length
- E Young's modulus
- ρ material density
- u_1 axial displacement
- x_1 axial coordinate

Objectives:

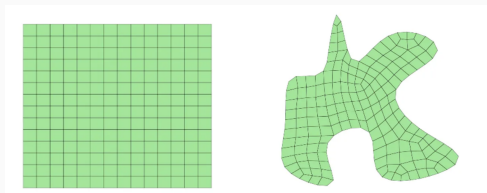
- ① Use the finite element method with n linear shape functions to approximate the first two natural frequencies of the bar.
- ② Compare the results with those obtained from Galerkin's approximation.

Localization and elementary quantities

Drawbacks of the global approach

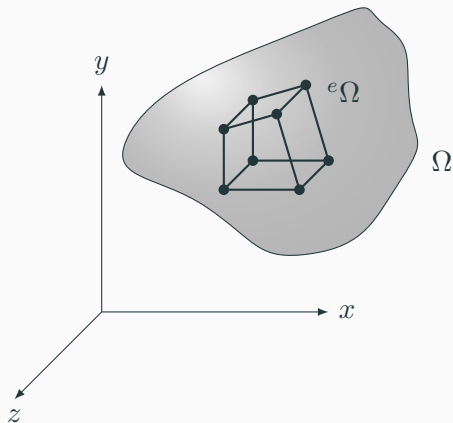
- ✗ Limited capability in handling complex (unstructured) mesh topologies.
- ✗ Computationally expensive: it requires defining one shape function per node.
- ✗ Limited utilization of the compact support of nodal shape functions.

Local approach: provides a quicker and more systematic way to compute the stiffness and mass matrices and the applied forces vector:



(Credit: Onscale - structured vs unstructured meshes)

Localization



- Let p be the number of nodes.
- Let m be the number of finite elements.
- Let ${}^e\Omega$ a finite element ($e = 1, \dots, m$).
- Let ep the number of nodes in the element ${}^e\Omega$.

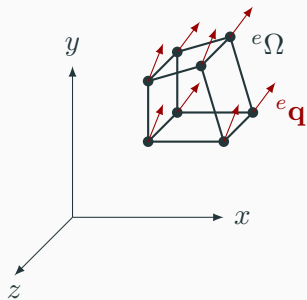
Localization of displacements

Restriction of displacements $\mathbf{u}^h$ and $\delta\mathbf{u}^h$ on the finite element ${}^e\Omega$:

$${}^e\mathbf{u}^h(\mathbf{x}, t) = {}^e\mathbf{H}(\mathbf{x}){}^e\mathbf{q}(t) \quad {}^e\delta\mathbf{u}^h(\mathbf{x}) = {}^e\mathbf{H}(\mathbf{x}){}^e\delta\mathbf{q}$$

- ${}^e\mathbf{u}^h$ restriction (3×1) of the displacement vector $\mathbf{u}^h$ on the finite element ${}^e\Omega$.
- ${}^e\delta\mathbf{u}^h$ restriction (3×1) of the virtual displacement vector $\delta\mathbf{u}^h$ on the finite element ${}^e\Omega$.
- ${}^e\mathbf{H}$ matrix (3×3^ep) of elementary shape functions of the finite element ${}^e\Omega$.
- ${}^e\mathbf{q}$ vector ($3^ep \times 1$) of unknown nodal displacements in the finite element ${}^e\Omega$.
- ${}^e\delta\mathbf{q}$ vector ($3^ep \times 1$) of nodal displacements in the finite element ${}^e\Omega$.

Local displacements approximation



$${}^e\mathbf{u}^h = \left[{}^eh_1\mathbf{I} \mid {}^eh_2\mathbf{I} \mid \dots \mid {}^eh_{e_p}\mathbf{I} \right] \begin{pmatrix} {}^e\mathbf{q}_1 \\ {}^e\mathbf{q}_2 \\ \vdots \\ {}^e\mathbf{q}_{e_p} \end{pmatrix}$$

$${}^e\delta\mathbf{u}^h = \left[{}^eh_1\mathbf{I} \mid {}^eh_2\mathbf{I} \mid \dots \mid {}^eh_{e_p}\mathbf{I} \right] \begin{pmatrix} {}^e\delta\mathbf{q}_1 \\ {}^e\delta\mathbf{q}_2 \\ \vdots \\ {}^e\delta\mathbf{q}_{e_p} \end{pmatrix}$$

Localisation matrices

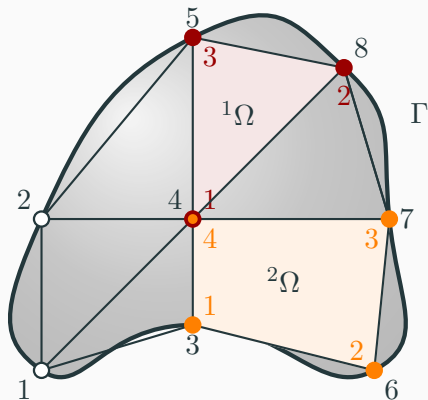
$$\begin{array}{c} \begin{array}{|c|} \hline {}^e\mathbf{q}_1 \\ \vdots \\ {}^e\mathbf{q}_{e_p} \\ \hline \end{array} \\ \begin{array}{c} \updownarrow \\ 3^e p \\ \updownarrow \end{array} \\ \begin{array}{c} \leftarrow 1 \rightarrow \end{array} \end{array} = \begin{array}{c} \begin{array}{|ccc|} \hline {}^e\mathbf{L}_{1,1} & \dots & {}^e\mathbf{L}_{1,p} \\ \vdots & & \vdots \\ {}^e\mathbf{L}_{e_p 1} & \dots & {}^e\mathbf{L}_{e_p,p} \\ \hline \end{array} \\ \begin{array}{c} \leftarrow 3p \rightarrow \end{array} \end{array} \cdot \begin{array}{c} \begin{array}{|c|} \hline \mathbf{q}_1 \\ \vdots \\ \mathbf{q}_p \\ \hline \end{array} \\ \begin{array}{c} \updownarrow \\ 3p \\ \updownarrow \end{array} \\ \begin{array}{c} \leftarrow 1 \rightarrow \end{array} \end{array}$$

$${}^e\mathbf{q} = {}^e\mathbf{L}\mathbf{q}$$

${}^e\mathbf{L}$ is a Boolean location matrix:

- ${}^e\mathbf{L}_{ij} = \mathbf{I}$ (3×3 identity matrix) if global node j corresponds to local node i ,
- ${}^e\mathbf{L}_{ij} = \mathbf{0}$ (3×3 null matrix) otherwise.

Localization matrices - example



- Number of nodes in the mesh: $p = 8$.
- Number of elements in the mesh: $m = 6$.
- Number of nodes in the element ${}^1\Omega$: ${}^1p = 3$.

$${}^1\mathbf{L} = \begin{bmatrix} 0 & 0 & 0 & \mathbf{I} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \mathbf{I} \\ 0 & 0 & 0 & 0 & \mathbf{I} & 0 & 0 & 0 \end{bmatrix}$$

- Number of nodes in the element ${}^2\Omega$: ${}^2p = 4$.

$${}^2\mathbf{L} = \begin{bmatrix} 0 & 0 & \mathbf{I} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \mathbf{I} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \mathbf{I} & 0 \\ 0 & 0 & 0 & \mathbf{I} & 0 & 0 & 0 & 0 \end{bmatrix}$$



Practical example:

Consider a mesh made of

- $p = 10'000$ nodes,
- trilinear hexahedral finite elements with $^e p = 8$ nodes each.

Every localization matrix $^e \mathbf{L}$ contains

- 720'000 entries,
- of which $3^e p = 24$ are 1s,
- the remaining 719'976 are 0s.

Connectivity table: local to global node numbering

- Elements and their connectivity are defined using a table.
- Example of connectivity table:

e_Ω	$^1\Omega$	$^2\Omega$	$^3\Omega$	$^4\Omega$
1	1	2	4	5
2	2	3	5	6
3	5	6	8	9
4	4	5	7	8

- The connectivity table provides the global numbering for each node in each element, corresponding to a column in the table above.
- The localization matrices $^e\mathbf{L}$ can be constructed from the connectivity table.

Additivity of integrals

- Approximated weak form:

$$\int_{\Omega} (\nabla \delta \mathbf{u}^h)^T \mathbf{C} \nabla \mathbf{u}^h d\Omega + \int_{\Omega} \rho (\delta \mathbf{u}^h)^T \ddot{\mathbf{u}}^h d\Omega = \int_{\Gamma_{\sigma}} (\delta \mathbf{u}^h)^T \hat{\mathbf{f}} d\Gamma + \int_{\Omega} (\delta \mathbf{u}^h)^T \mathbf{f} d\Omega$$

- We localize the integrals using their additivity property:

$$\begin{aligned} & \sum_{e=1}^m \left(\int_{e\Omega} (\nabla^e \delta \mathbf{u}^h)^T \mathbf{C} \nabla^e \mathbf{u}^h d\Omega + \int_{e\Omega} \rho (^e \delta \mathbf{u}^h)^T {}^e \ddot{\mathbf{u}}^h d\Omega \right) \\ &= \sum_{e=1}^m \left(\int_{e\Gamma_{\sigma}} (^e \delta \mathbf{u}^h)^T \hat{\mathbf{f}} d\Gamma + \int_{e\Omega} (^e \delta \mathbf{u}^h)^T \mathbf{f} d\Omega \right) \end{aligned}$$

and consider the local quantities ${}^e \mathbf{u}^h = {}^e \mathbf{H}^e \mathbf{L} \mathbf{q}$ and ${}^e \delta \mathbf{u}^h = {}^e \mathbf{H}^e \mathbf{L} \delta \mathbf{q}$.

Additivity of integrals - inertial forces

Recall ${}^e\mathbf{u}^h = {}^e\mathbf{H}^e\mathbf{L}\mathbf{q}$ and ${}^e\delta\mathbf{u}^h = {}^e\mathbf{H}^e\mathbf{L}\delta\mathbf{q}$.

- 1 Consider only the term related to the virtual work of inertial forces (acceleration):

$$\sum_{e=1}^m \int_{{}^e\Omega} \rho ({}^e\delta\mathbf{u}^h)^T {}^e\ddot{\mathbf{u}}^h d\Omega = \delta\mathbf{q}^T \left[\sum_{e=1}^m {}^e\mathbf{L}^T \underbrace{\left(\int_{{}^e\Omega} \rho {}^e\mathbf{H}^T {}^e\mathbf{H} d\Omega \right)}_{{}^e\mathbf{M}} {}^e\mathbf{L} \right] \ddot{\mathbf{q}}$$

Additivity of integrals - internal and external forces

Recall ${}^e\mathbf{u}^h = {}^e\mathbf{H}^e\mathbf{L}\mathbf{q}$ and ${}^e\delta\mathbf{u}^h = {}^e\mathbf{H}^e\mathbf{L}\delta\mathbf{q}$.

- ② Analogously for the term related to the virtual work of internal forces:

$$\sum_{e=1}^m \int_{{}^e\Omega} ({}^e\nabla {}^e\delta\mathbf{u}^h)^T \mathbf{C} {}^e\nabla {}^e\mathbf{u}^h d\Omega = \delta\mathbf{q}^T \left[\sum_{e=1}^m {}^e\mathbf{L}^T \underbrace{\left(\int_{{}^e\Omega} {}^e\nabla \mathbf{H}^T \mathbf{C} {}^e\nabla \mathbf{H} d\Omega \right)}_{{}^e\mathbf{K}} {}^e\mathbf{L} \right] \mathbf{q}$$

- ③ Term related to the virtual work of external forces:

$$\begin{aligned} & \sum_{e=1}^m \int_{{}^e\Gamma_\sigma} ({}^e\delta\mathbf{u}^h)^T \hat{\mathbf{f}} d\Gamma + \int_{{}^e\Omega} ({}^e\delta\mathbf{u}^h)^T \mathbf{f} d\Omega \\ &= \delta\mathbf{q}^T \sum_{e=1}^m {}^e\mathbf{L}^T \underbrace{\left(\int_{{}^e\Gamma_\sigma} {}^e\mathbf{H}^T \hat{\mathbf{f}} d\Gamma + \int_{{}^e\Omega} {}^e\mathbf{H}^T \mathbf{f} d\Omega \right)}_{{}^e\mathbf{r}(t)} \end{aligned}$$

- **Elementary stiffness matrix** ($3^ep \times 3^ep$):

$${}^e\mathbf{K} = \int_{{}^e\Omega} {}^e\mathbf{B}^T \mathbf{C}^e \mathbf{B} \, d\Omega$$

$${}^e\mathbf{B} = \nabla {}^e\mathbf{H} = \left[\begin{array}{c|c|c} \nabla {}^eh_1 & \dots & \nabla {}^eh_{e_p} \end{array} \right] \text{ elementary deformation matrix } (6 \times 3^ep)$$

.

- **Elementary mass matrix** ($3^ep \times 3^ep$):

$${}^e\mathbf{M} = \int_{{}^e\Omega} \rho {}^e\mathbf{H}^T {}^e\mathbf{H} \, d\Omega.$$

- **Elementary applied forces vector** ($3^ep \times 1$):

$${}^e\mathbf{r}(t) = \int_{{}^e\Gamma_\sigma} {}^e\mathbf{H}^T \hat{\mathbf{f}} \, d\Gamma + \int_{{}^e\Omega} {}^e\mathbf{H}^T \mathbf{f} \, d\Omega.$$

Assembly operator

We define the assembly operator as follows:

$$\mathbf{K} = \bigcup_{e=1}^m {}^e\mathbf{K} = \sum_{e=1}^m {}^e\mathbf{L}^T {}^e\mathbf{K} {}^e\mathbf{L}$$

$$\mathbf{M} = \bigcup_{e=1}^m {}^e\mathbf{M} = \sum_{e=1}^m {}^e\mathbf{L}^T {}^e\mathbf{M} {}^e\mathbf{L}$$

$$\mathbf{r} = \bigcup_{e=1}^m {}^e\mathbf{r} = \sum_{e=1}^m {}^e\mathbf{L}^T {}^e\mathbf{r}$$

Automating integration and archetypal shape functions

Physical structure:

Ω



Elements:

$^1\Omega, ^2\Omega, \dots, ^m\Omega$



Master elements:

$\Omega^a, \Omega^b, \dots$



To automate the integration and simplify the definition of shape functions, we transform each *distorted* finite element ${}^e\Omega$ into a **reference (archetypal or master) element** Ω^a where we can apply standard numerical integration schemes.

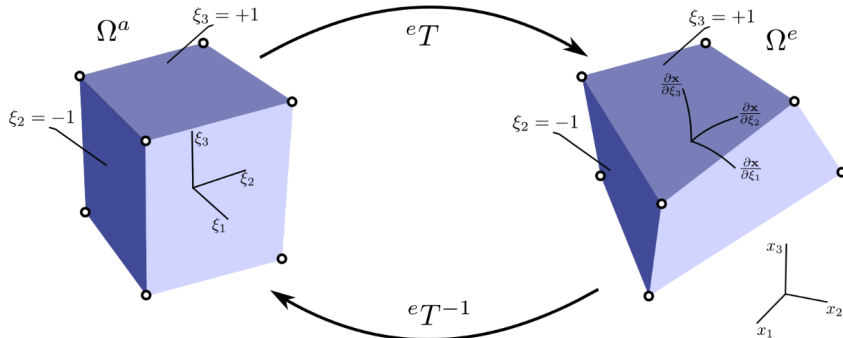
- The coordinate transformation:

$$\begin{aligned} {}^eT : \Omega^a &\rightarrow {}^e\Omega \\ \boldsymbol{\xi} &\mapsto \mathbf{x}(\boldsymbol{\xi}) \end{aligned}$$

maps any point $\boldsymbol{\xi} = \{\xi_1, \xi_2, \xi_3\}^T$ in Ω^a to its corresponding point of coordinate $\mathbf{x} = \mathbf{x}(\boldsymbol{\xi}) = \{x_1(\boldsymbol{\xi}), x_2(\boldsymbol{\xi}), x_3(\boldsymbol{\xi})\}^T$ in ${}^e\Omega$.

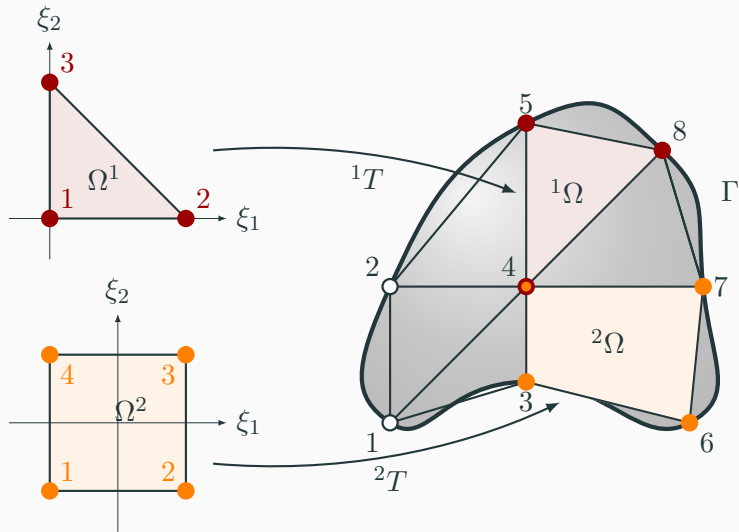
- eT is a bijective application.

An illustration of a coordinate transform eT in 3D



(Credit: Joel Cugnoni - Finite Element Method applied to linear statics of deformable solids)

An illustration of coordinate transforms in 2D



Biunivocity of coordinate transformation eT

$${}^e\mathbf{J} = \begin{bmatrix} \frac{\partial x_1}{\partial \xi_1} & \frac{\partial x_1}{\partial \xi_2} & \frac{\partial x_1}{\partial \xi_3} \\ \frac{\partial x_2}{\partial \xi_1} & \frac{\partial x_2}{\partial \xi_2} & \frac{\partial x_2}{\partial \xi_3} \\ \frac{\partial x_3}{\partial \xi_1} & \frac{\partial x_3}{\partial \xi_2} & \frac{\partial x_3}{\partial \xi_3} \end{bmatrix} \quad {}^e\mathbf{J}^{-1} = \begin{bmatrix} \frac{\partial \xi_1}{\partial x_1} & \frac{\partial \xi_1}{\partial x_2} & \frac{\partial \xi_1}{\partial x_3} \\ \frac{\partial \xi_2}{\partial x_1} & \frac{\partial \xi_2}{\partial x_2} & \frac{\partial \xi_2}{\partial x_3} \\ \frac{\partial \xi_3}{\partial x_1} & \frac{\partial \xi_3}{\partial x_2} & \frac{\partial \xi_3}{\partial x_3} \end{bmatrix}$$

- ${}^e\mathbf{J}$ is the Jacobian matrix associated with eT : ${}^e\mathbf{J}_{ij} = \frac{\partial x_i}{\partial \xi_j}$ ($i, j = 1, 2, 3$),
- ${}^e\mathbf{J}^{-1}$ is the inverse Jacobian matrix,
- ${}^ej = \det({}^e\mathbf{J})$ is the determinant of the Jacobian matrix ${}^e\mathbf{J}$.

Sufficient condition for invertibility: if ${}^ej > 0$ everywhere in ${}^e\Omega$, then eT is invertible in ${}^e\Omega$ and ${}^e\mathbf{J}^{-1}$ exists.

Master elements and master shape functions

- The coordinate transformation ${}^eT^{-1}$ maps shape functions on ${}^e\Omega$ to the master element space Ω^a :

$${}^a\mathbf{H}(\boldsymbol{\xi}) = {}^e\mathbf{H}(\mathbf{x}(\boldsymbol{\xi}))$$

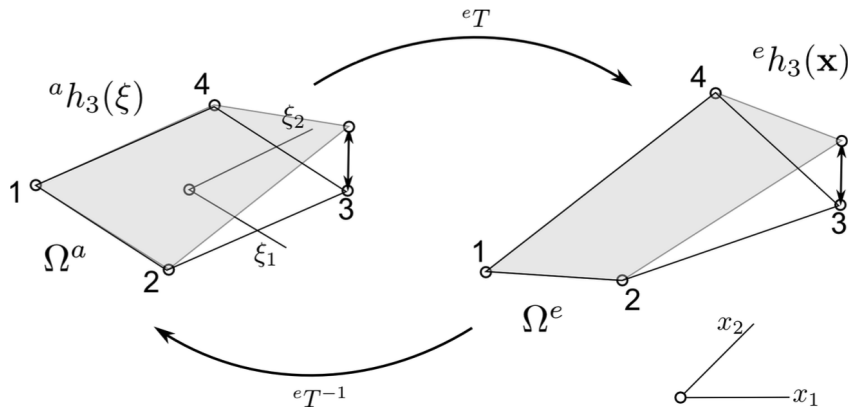
- Inside each element ${}^e\Omega$:

$${}^e\mathbf{u}^h[\mathbf{x}(\boldsymbol{\xi})] = {}^a\mathbf{H}(\boldsymbol{\xi}){}^e\mathbf{q}$$

- This allows shape function ${}^a\mathbf{H}(\boldsymbol{\xi})$ to be defined only *once* on each master element ${}^a\Omega$.

 **Result:** master shape functions!

An illustration of a master shape function



(Credit: Joel Cugnoni - Finite Element Method applied to linear statics of deformable solids)

Choosing a simple coordinate transformation

👉 Coordinate transformations are defined using the shape functions:

$${}^eT : \mathbf{x}(\boldsymbol{\xi}) = {}^a\mathbf{H}(\boldsymbol{\xi}){}^e\mathbf{x} = \sum_{i=1}^{e_p} {}^ah_i(\boldsymbol{\xi}){}^e\mathbf{x}_i$$

- Inside each element ${}^e\Omega$, the local coordinates are interpolated as a linear combination of master shape functions ah_i and nodal coordinates ${}^e\mathbf{x}_i$.
- Kronecker property ensures node correspondence:

$${}^eh_i(\mathbf{x}_j) = \delta_{ij} \quad \Rightarrow \quad {}^ah_i(\boldsymbol{\xi}_j) = \delta_{ij}.$$

- This guarantees that each node of the master element Ω^a maps to a corresponding node in the deformed element ${}^e\Omega$.

Integration by substitution formulas

- Given ${}^eT : \Omega^a \rightarrow {}^e\Omega$, an integral over ${}^e\Omega$ of a function $F : {}^e\Omega \rightarrow \mathbb{R}$ can be rewritten as an integral over Ω^a :

$$\int_{{}^e\Omega} F(\mathbf{x}) d\mathbf{x} = \int_{\Omega^a} F(\mathbf{x}(\boldsymbol{\xi})) {}^eJ d\boldsymbol{\xi}$$

- When the integrand involves the operator ∇ , then:

$$\int_{{}^e\Omega} \nabla_{\mathbf{x}} F(\mathbf{x}) d\mathbf{x} = \int_{\Omega^a} \nabla_{\boldsymbol{\xi}} F(\mathbf{x}(\boldsymbol{\xi})) {}^e\mathbf{J}^{-1} {}^eJ d\boldsymbol{\xi}$$

since

$$\frac{\partial}{\partial x_i} = \sum_{j=1}^3 \frac{\partial \xi_j}{\partial x_i} \frac{\partial}{\partial \xi_j} = \sum_{j=1}^3 {}^e\mathbf{J}_{ij}^{-1} \frac{\partial}{\partial \xi_j}.$$

Master elements and derivatives

- The spatial derivative operator $\nabla_{\mathbf{x}}$ is defined in the global coordinate system (x_1, x_2, x_3) . Applying the coordinate transform ${}^eT^{-1}$, we can then extend it to be applied on the master element Ω^a :

$$\frac{\partial}{\partial x_i} = \sum_{j=1}^3 \frac{\partial \xi_j}{\partial x_i} \frac{\partial}{\partial \xi_j} = \sum_{j=1}^3 {}^e\mathbf{J}_{ij}^{-1} \frac{\partial}{\partial \xi_j}.$$

- The elementary strain-displacement matrix ${}^e\mathbf{B}$ can be directly derived from the master shape functions ${}^a\mathbf{H}$:

$${}^e\mathbf{B} = \left[\nabla_{\mathbf{x}} {}^e h_1 \mid \dots \mid \nabla_{\mathbf{x}} {}^e h_{e_p} \right] = \left[\nabla_{\boldsymbol{\xi}} {}^a h_1 {}^e\mathbf{J}^{-1} \mid \dots \mid \nabla_{\boldsymbol{\xi}} {}^a h_{e_p} {}^e\mathbf{J}^{-1} \right].$$

Automating the integration

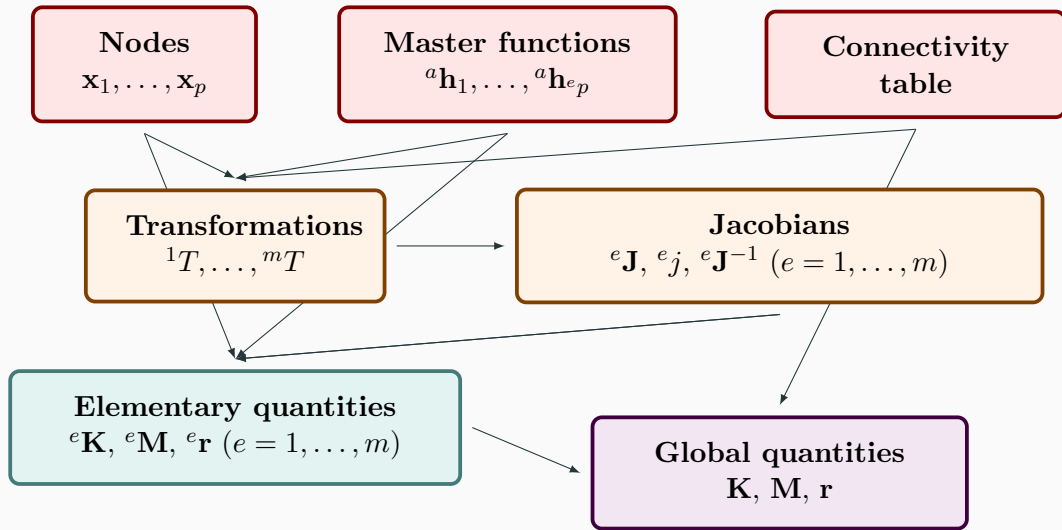
Using the coordinate transform eT and master shape functions, the integrals in the definitions of ${}^e\mathbf{K}$, ${}^e\mathbf{M}$ and ${}^e\mathbf{r}$, can be carried out directly on a standard domain Ω^a :

$${}^e\mathbf{K} = \int_{\Omega^a} (\nabla_{\xi} {}^a\mathbf{H}^e \mathbf{J}^{-1})^T \mathbf{C} (\nabla_{\xi} {}^a\mathbf{H}^e \mathbf{J}^{-1}) {}^e j d\xi$$

$${}^e\mathbf{M} = \int_{\Omega^a} \rho {}^a\mathbf{H}^{Ta} \mathbf{H} {}^e j d\xi$$

$${}^e\mathbf{r}(t) = \int_{\Gamma_{\sigma}^a} {}^a\mathbf{H}^T \hat{\mathbf{f}} {}^e j|_{\Gamma_{\sigma}^a} d\Gamma + \int_{\Omega^a} {}^a\mathbf{H}^T \mathbf{f} {}^e j d\xi$$

Systematization of the finite elements algorithm



Isoparametric, subparametric, superparametric elements

Displacement field as well as the geometrical representation of the finite elements could be approximated using different sets of shape functions.

■ Geometrical representation

$${}^eT : \mathbf{x}(\boldsymbol{\xi}) = {}^a\mathbf{H}(\boldsymbol{\xi}) {}^e\mathbf{x}$$

■ Displacement field approximation

$${}^e\mathbf{u}^h(\mathbf{x}, t) = {}^a\mathbf{H}(\boldsymbol{\xi}(\mathbf{x})) {}^e\mathbf{q}(t)$$

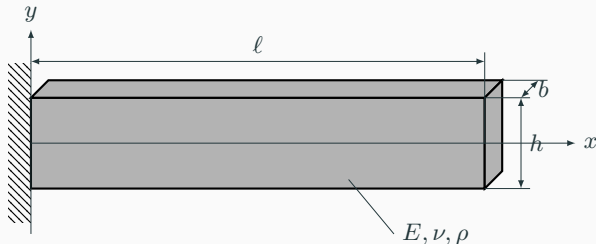
- ① Subparametric: less nodes for geometric than for displacement,
- ② Isoparametric: same nodes for both geometry and displacement,
- ③ Superparametric: more nodes for geometric than for displacement.

Example: modal analysis of a clamped beam

Modal analysis of a clamped beam

Kinematic assumptions:

- The beam is made of elastic material which is homogeneous and isotropic (E , ν and ρ).
- Assume a plane stress state (very small thickness). The structure can be modeled in the (x, y) plane using two-dimensional finite elements.



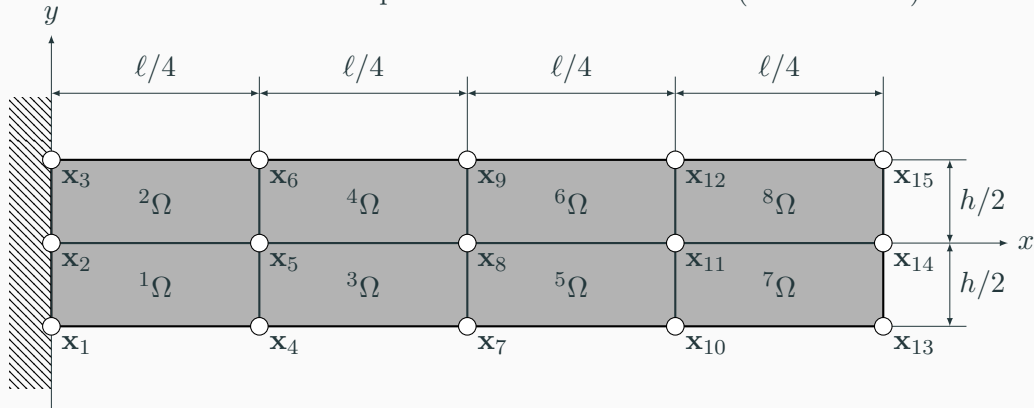
- $E = 210 \text{ GPa}$ Young's modulus
- $\nu = 0.3$ Poisson's ratio
- $\rho = 7800 \text{ kg/m}^3$ material density
- $\ell = 2 \text{ m}$ length
- $h = 0.5 \text{ m}$ height
- $b = 0.01 \text{ m}$ thickness
- x axial coordinate
- y transversal coordinate

Variables:

- $u_1(x, y, t)$ axial displacement
- $u_2(x, y, t)$ transversal displacement

Modal analysis of a clamped beam

Discretization into 8 bilinear quadrilateral finite elements (4 nodes each)



Objective: determine the lower natural frequencies of the beam.